

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 1.3: Quadratic Equations — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.3: Quadratic Equations
Driving Question	DRIVING QUESTION: How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Forming Algebraic Expressions	Achieng, a student volunteer at a Kisumu prize-giving ceremony, could not find whole-number rows and columns for 120 chairs (columns exceed rows by 7) using guess-and-check or simple division. Across 5 Kenyan contexts — from Rift Valley maize farms to Kongowea Market tomato displays to Kericho fabric pieces — the same pattern appeared: two related unknowns whose product is known.	When one unknown quantity exceeds another by a fixed amount and their product is known, we can represent the smaller unknown with a letter (e.g., r), write the larger as an expression (e.g., $r + 7$), and form a product equation ($r(r + 7) = 120$). Expanding and rearranging gives a quadratic equation — $r^2 + 7r - 120 = 0$ — where the unknown appears squared. This structure cannot be solved by simple division because both unknowns are linked.	A quadratic equation arises naturally whenever two related linear unknowns multiply to give a fixed total. The squared term appears because we multiply r by itself (indirectly, through expanding $r(r+7)$). Algebra gives us a precise language to write the relationship — turning Achieng's seating puzzle into $r^2 + 7r - 120 = 0$ — even before we know how to solve it. Solving this equation (finding the value of r) is the central challenge for the rest of the sub-strand.
Lesson 2 From Expressions to Quadratic Expressions: Recognising and Writing $ax^2 + bx + c$	When we multiplied the number of rows (x) by the number of columns ($x + 7$) for the Kisumu chair arrangement, and when we multiplied the sides of the Eldoret maize plot (x by $x + 5$) and the Kericho mandazi tray ($(x + 3)$ by $(x - 2)$), every product produced an expression with an x^2 term as the highest power.	Multiplying two linear expressions always produces a quadratic expression of the form $ax^2 + bx + c$, where $a \neq 0$. The x^2 term appears because $x \times x = x^2$. We can identify a (coefficient of x^2), b (coefficient of x), and c (the constant term) in any expanded quadratic expression. A quadratic expression differs from a linear expression because its highest power of x is 2, not 1.	The reason the Kisumu seating problem — and any rectangular-arrangement or area problem where one dimension exceeds the other by a fixed number — naturally produces a quadratic expression is that area (or total count) equals length $\times$ width, and when both dimensions depend on the same unknown x , their product must contain x^2 . This is why Achieng's teacher Mr Otieno said the relationship involves a squared unknown: it is unavoidable once the two dimensions are expressed in terms of the same variable.

Lesson 3 Forming Quadratic Equations from Real-Life Situations	Learners read three Kenyan real-life scenario cards (Rift Valley fencing, Kisumu prize-giving seating, Kericho seedling grid) and recorded the numerical relationships described in each — including the total area or total count and the way the two dimensions were related.	Learners learned that when two related unknown quantities are multiplied together to give a known total, the resulting equation always contains a squared term (degree 2), making it a quadratic equation. They identified three defining features of a quadratic equation: exactly one squared term, a non-zero coefficient for that term, and the equation set equal to zero. They also named three common formation errors: the linear error, the cancellation error, and the incomplete-form error.	Learners explained why Achieng's seating problem — and the other two scenarios — cannot be described by a simple linear equation: because the two dimensions are both expressed in terms of the same unknown, their product inevitably produces a squared term. They annotated their cumulative seating-layout model with the Let statement, the product equation, and the standard quadratic form $r^2 + 7r - 120 = 0$, and wrote a Model Status Note acknowledging that forming the equation is now within reach but solving it requires a new method.
Lesson 4 Factorisation of Quadratic Expressions: The Product-and-Sum Method	Learners observed a factor-pair table for the expression $n^2 + 3n - 108$ (Kericho maize seedling arrangement), completing the sum column for all integer factor pairs of 108 and identifying the pair $(-9, +12)$ whose product is -108 and sum is $+3$.	Learners learned the product-and-sum method for factorising quadratic expressions: identify two numbers whose product equals the constant term and whose sum equals the coefficient of the linear term, rewrite the middle term using these two numbers, group and factorise, then verify by expanding. They applied this method to Achieng's chair-arrangement expression $n^2 + 7n - 120 = (n + 15)(n - 8)$.	Learners explained that factorisation reverses expansion and that every factorisation can be verified by multiplying the brackets back out. They updated the DQB with the open question 'Can factorisation always work?', and annotated their cumulative seating-arrangement model with the factorised form and product-and-sum working, noting that the equation must still be set equal to zero and solved — the goal of Lesson 5.
Lesson 5 Solving Quadratic Equations by Factorisation: Finding and Interpreting Both Roots	We observed that the equation $x^2 + 7x - 120 = 0$ (Achieng's chair problem) and the Kericho farm equation both produced TWO solutions when solved by factorisation.	We learned that the Zero Product Property lets us set each factor equal to zero to find both roots, and that we must use the real-life context to decide which root(s) are valid — rejecting negative values where dimensions or counts cannot be negative.	We explained why Achieng's answer is 8 rows and 15 columns (not -15 rows): the algebra gives two mathematically correct roots, but only the positive root makes sense for a physical seating arrangement. This is the 'Context Filter' step in our growing model.
Lesson 6 Verification and Deeper Factorisation Practice — Checking Solutions and Contextual Validity	Learners observed that substituting $r = 8$ into $r^2 + 7r - 120 = 0$ gives $64 + 56 - 120 = 0$ ✓, and substituting $r = -15$ gives $225 - 105 - 120 = 0$ ✓ — both roots satisfy the equation algebraically. They also observed that in the Kisumu market-trader problem, one root was negative (contextually impossible as a physical width), while in the Iken athletics problem both roots were non-negative and both could be physically meaningful.	Verifying a solution means substituting it back into the original equation and confirming the result equals zero — this is a separate, obligatory step after factorisation. A quadratic equation always yields two algebraic roots, but the real-life context determines how many are valid: sometimes only one, sometimes both, and (in cases not yet encountered) possibly neither. The three-question Validity Decision Rule — checking sign constraints, whole-number constraints, and physical possibility — is the tool for making this judgement	Achieng's original puzzle is now fully resolved: both $r = 8$ and $r = -15$ are genuine solutions to $r^2 + 7r - 120 = 0$, which is why the equation is correct and why 'guessing and checking' failed her. However, only $r = 8$ is valid for the chair arrangement at the Kisumu prize-giving ceremony, because the number of rows cannot be a negative number. The 8 rows and 15 columns ($8 + 7 = 15$) give exactly $8 \times 15 = 120$ seats, confirming the mathematical model matches the real-world situation perfectly.

		systematically.	
Lesson 7 Applications and Digital Research: Quadratic Equations in the Real World	Learners researched and identified real-world examples of quadratic equations in architecture (e.g., rectangular floor plans and arch spans in Nairobi), agriculture (e.g., rectangular maize shamba dimensions in the Rift Valley), sports (e.g., football trajectory at Kasarani Stadium), and business (e.g., a Mombasa market trader's profit function). They observed that in each case, a product of two related linear expressions — equal to a known total — generates a quadratic equation of the form $ax^2 + bx + c = 0$.	Learners learned that: (i) the quadratic structure first seen in Achieng's chair problem (rows $\times$ columns = 120, where columns exceed rows by 7) recurs across many real-life domains in Kenya; (ii) solving a quadratic equation produces two mathematical solutions, but real-life context (no negative lengths, areas, times, or quantities) determines which solution is physically valid; (iii) observing online safety measures is a responsibility when using digital devices for mathematical research; and (iv) quadratic equations are a tool for decision-making — in design, farming, sport, and commerce.	Learners explained why a squared unknown appears in real-life problems: whenever two related unknown quantities are multiplied together to give a known area, total, or quantity, the relationship produces a quadratic equation. They also explained the solution-validity rule: after solving by factorisation, substitute both solutions back into the real-life context and reject any solution that contradicts physical constraints (e.g., you cannot have a negative number of rows of chairs, a negative field width, or a negative time). This rule was connected explicitly to Achieng's rejection of $x = -15$ in Lesson 1's phenomenon.
Lesson 8 Model Building, Consolidation & Final Explanation: The Full Journey of a Quadratic Equation	Students examined an Evidence Wall of six stations tracing the complete unit journey — from Achieng's 120-chair seating problem at a Kisumu school to the factorised solution and real-life interpretation. They identified which solution ($x = 8$ rows) is valid and why $x = -15$ must be rejected.	A quadratic equation of the form $x^2 + 7x - 120 = 0$ arises when two related unknown dimensions (rows and columns) have a known product. It is solved by finding factor pairs of the constant term whose sum equals the coefficient of x . Both solutions must be tested in context — negative values are rejected because physical quantities such as rows and columns cannot be negative.	Learners constructed cumulative journey models mapping the full path from real-life context $\rightarrow$ algebraic expression $\rightarrow$ quadratic equation $\rightarrow$ factorisation $\rightarrow$ solution $\rightarrow$ contextual interpretation. They presented these models, closed the Driving Question Board, completed an individual written assessment task (Kericho shamba dimensions), and connected quadratic equations to careers across Kenya including architecture, agriculture, event management, and fisheries.

ATTRIBUTION AND LICENSING

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